

Gioia Zheng

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github.com/GioiaZheng

Profile

Bachelor's student in Applied Computer Science and Artificial Intelligence at Sapienza University of Rome. Focused on **information retrieval**, **retrieval-augmented generation (RAG)**, and **reproducible ML evaluation**.

Education

Sapienza University of Rome

Sep 2022 – Present

B.Sc. in Applied Computer Science and Artificial Intelligence

Experience

NLP Algorithm Engineer Intern

Feb 2026 – May 2026

Microsoft — Natural Language Computing (Remote)

- Built an end-to-end retrieval-augmented QA pipeline on MS MARCO: BM25 + dense (SBERT/FAISS) retrieval, cross-encoder reranking, T5 generation, paired-bootstrap evaluation
- Cross-encoder reranking of dense top-100 doubled generation Token-F1 (0.197 → 0.368)** on full dev/small (6,980 queries); 95% paired-bootstrap CIs on all four surface-form Δ s strictly above 0
- DistilBERT BERTScore proxy on 3,000 paired examples confirmed the gain is not a surface-form artefact ($\Delta +0.173$); a deterministic failure taxonomy traced $\sim 90\%$ of regressions to generator-side truncation, not retrieval failure
- Surfaced an NLI-entailment Δ sign reversal (-0.145) — the only metric in the pipeline whose paired Δ reverses sign — motivating a generator-capacity follow-up rather than a decode-budget fix
- Reproducibility scaffolding: per-run manifest (git SHA, config hash, dependency-file hashes, env fingerprint), lockfile, 286 unit tests, CI on every push

Web Development Intern

Mar 2026 – Present

Idearia, Rome, Italy

- Developed and maintained PHP/WordPress backend systems for client-facing websites
- Designed n8n automation workflows to integrate WordPress content and forms with external services

Programming Intern

May 2021 – Jun 2021

Sandbox S.r.l., Italy

- Contributed to a 3D simulation platform and backend tooling

Projects

Full-Stack Chat System

github.com/GioiaZheng/go-chat-system

- Full-stack chat application; Go backend with concurrent request handling and SQLite persistence
- REST APIs documented in OpenAPI; Vue.js frontend embedded into the Go binary at build time for single-artifact deployment

CiboCompass Mobile App

github.com/GioiaZheng/CiboCompass

- Cross-platform React Native app helping international students navigate Italian restaurant menus
- Offline-first rating system with local cache and backend sync; Go REST API with SQLite persistence

Handwritten OCR System

github.com/GioiaZheng/handwritten-ocr-system

- End-to-end handwritten OCR pipeline in PyTorch on the IAM handwriting dataset
- CNN feature extractor with bidirectional LSTM and CTC alignment for variable-length transcription; evaluated with CER and WER on held-out splits

Skills

Programming

Python, Go, JavaScript, PHP, SQL

ML & NLP

PyTorch, Hugging Face Transformers, RAG, Deep Learning

Information Retrieval

BM25, FAISS, sparse + dense retrieval, cross-encoder reranking, BERTScore, NLI grounding, paired-bootstrap evaluation

Backend & Systems Tools

REST APIs, OpenAPI, SQLite, concurrent backend (Go), system design

Languages

Git, Docker, Linux, GitHub Actions

English (Professional), Italian (Bilingual), Chinese (Native), German (Beginner)

Additional

WISWUP — 1st Place

Sep 2021